

Article

Enriching Building Information Modeling Models through Information Delivery Specification

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Abstract: The efficient acquisition and dissemination of information are crucial in building information modeling (BIM). Current BIM models face significant challenges, including inadequate modeling techniques, poorly defined information requirements, and low interoperability. These issues result in poor information quality and complicate the transition from information acquisition to model processing. Public authorities often provide documentation in various formats, requiring manual transfer to software, which is error-prone and burdensome. This process is particularly difficult for small and medium enterprises lacking resources and knowledge. To address these issues, the IDS (Information Delivery Specification) Collab Tool is under development. This tool aims to automate the import of requirements into authoring software, perform automated compliance checks, and enhance interoperability among stakeholders. It will assist designers in providing accurate information according to requirements through the IDS standard, improving model quality and efficiency from early design stages. Adapting BIM models to specific project requirements and aligning new IDS capabilities with traditional industry practices remain significant challenges. Preliminary evaluations indicate the tool's potential to significantly improve workflow efficiency and compliance in BIM modeling. However, broader awareness and adoption of the IDS standard are needed. Further research and refinement are essential to fully realize the benefits of digital tools in revolutionizing design and construction practices.

Keywords: information delivery specification; building information modeling; information requirement; industry foundation classes; automated compliance checking



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1. Introduction

Within the field of digital construction, especially in small and medium enterprises (SMEs), a transformation in the approach to design and construction has started [1–4]. This research project focused on a specific use case in the Autonomous Province of Bolzano, South Tyrol (Italy). It brought insightful knowledge of the role of public administrations and the digitalization process, thus leading to research-feasible applications of the most recent openBIM standards available. The IDS is a standard developed by buildingSMART International with significant potential for application in a wide range of fields. Kremer et al. [1] propose extending the scope of the IDS to incorporate additional verification resources, while Siddiqui et al. [2] discussed its use in occupant movement analysis (OMA) and Gragnaniello et al. [3] demonstrate its effectiveness in verifying the consistency of the exchange requirements (ERs). The IDS is employed to assign value and structure to the information that has been gathered. As the standard itself is relatively new, it is not currently being fully utilized and is relatively unknown to companies. This paper explores the role of the IDS beyond its acknowledged application for model validation [2–7], with particular emphasis on its significance in enhancing the design phase of construction projects. The essence of this discussion lies in the recognition that information becomes valuable knowledge when it is efficiently gathered and processed. Currently, the process of moving from

the acquisition of information to its processing in models is characterized by a high degree of intricacy and ineffectiveness, which results in low information interoperability [8–10]. Conventional approaches, which primarily use the IDS toward the end of project development for the validation of specific data [11], fail to capitalize on significant opportunities to optimize resources during the early design stages. By advocating for the early integration of the IDS into the design process, a strategy is proposed to effectively reduce the amount of time dedicated to modeling IRs. This approach is designed to speed up the design phase and also enhance the quality of the model by including accurate information from the beginning. The automation of consistent sharing of exchange requirements through the IDS is expected to support both design firms and public authorities (PAs) in dealing with better models.

The aim of the study was to develop a tool to automate the import of information requirements into authoring software and enhance interoperability in BIM processes. The objective was to improve model quality and efficiency from the early design stages, ultimately contributing to reducing errors, enhancing workflow efficiency, and ensuring compliance with standardized information requirements. The contribution of this study lies in providing a structured framework that bridges the gap between open standards and practical implementation, supporting both design firms and public authorities in managing better-designed projects.

2. Literature Review

The following subsections will provide a structured overview of concepts as a background for understanding the objectives. This literature review is mainly structured in five parts. The initial subsection provides an overview of automated compliance checking (ACC) in the IDS, which is followed by a discussion of the challenges encountered in research for information quality. The third subsection presents the integration of the IDS for enhanced interoperability, which can anticipate the automated checking compliance in the design stage, described in the fourth subsection. The final section of this literature review delves into the specification of information requirements and other openBIM solutions.

The integration of ACC and the IDS represents a pivotal advancement in the digital construction industry. It has the potential to address a number of key challenges, including those related to information exchange, interoperability, openBIM principles, and user requirement definition. This is due to its duality (human- and machine-readable) and vendor-neutral nature. These elements collectively enhance BIM modeling and automation tools, creating a cohesive software platform that supports the efficient execution of projects. Semantic Web technologies enhance ACC by using SHACL for human- and machine-readable constraints, aligning with openBIM principles and the IDS [12].

2.1. Background on ACC in IDS

ACC plays a critical role in ensuring that construction projects adhere to established standards and regulations. However, traditional ACC processes are often fragmented and heavily rely on software vendor-specific solutions. Since the first appearance of ACC, it was limited in its effectiveness and interoperability; nevertheless, a reliable framework was laid for research to come, as in Eastman et al.'s study [13], where the rule-checking process was outlined. It consisted of four main steps: 1. rule interpretation, 2. building model preparation, 3. rule execution, and, lastly, 4. rule check reporting.

To address the above-mentioned challenges, a comprehensive approach to ACC requires the adoption of open standards that can be universally applied across different software platforms and project stakeholders.

The IDS, a standard developed by buildingSMART International, offers a solution to these challenges by providing a structured framework for specifying and exchanging information requirements (IRs). The IDS facilitates an openBIM approach, enhancing interoperability among stakeholders at various levels, whether national or municipal, due to its semantic nature checks. Pauwels et al. demonstrated, through use cases, the automatic

validation of 66% of human-readable requirements, combining semantic and 3D geometric compliance checks [14]. Pauwels' results demonstrated the potential of semantic rule checking to enhance consistency and efficiency in validating project requirements.

2.2. Challenges in Information Quality

One of the major issues in the digital construction industry is the inadequacy of modeling techniques and the lack of clearly defined IRs, which results in issues with information quality (IQ). In facility management, for instance, Zadeh et al. [15] developed an assessment to evaluate the conformance of BIM to owner-specific IR. The metrics included completeness, accuracy, understandability, unambiguity, and well-formedness. His study resulted in three main findings: 1. essential information types, 2. attributes and characteristics, and 3. level of detail. Analogously, Leygonie et al. [16] addressed the operation and maintenance phase by developing a detailed checklist identifying necessary information. The study identified key challenges such as the lack of comprehensive guidelines for FM-BIM and the need for integrating automated tools into existing workflows, thus recommending leveraging artificial intelligence and machine learning to further automate quality control. As has been seen and observed by Tomczak et al. [17], the literature reveals a fragmented landscape when it comes to specifying IR. Tomczak et al. provided a comprehensive comparative analysis of various methods for defining IR, including but not limited to data dictionaries (ISO 12006), an Information Delivery Manual (ISO 29481), IFC property templates, the IDS, and level of information need (ISO 7817; still under publication). His conclusion states that while there is no one-size-fits-all solution for specifying IR in BIM, a conscious selection of methods based on project-specific needs and a clear understanding of each method's strengths and limitations can significantly enhance information management in digital construction projects. This results in challenges such as compliance with owner's specifications [10,18], interoperability [19,20], and modeling practices [21–23].

2.3. Integration of IDS for Improved Interoperability

Interoperability has been a longstanding issue in BIM workflows. Specifically, Siddiqui et al. address the challenges of inadequate modeling techniques and poorly defined information requirements, thus leading to a low grade of interoperability. Furthermore, they discuss the implementation of new BIM standards aimed at improving interoperability among stakeholders by ensuring that information is consistent and accessible [2]. Afzal et al. highlight the challenges of inadequate data transfer and integration between different modeling systems. Specifically, they address the issues of inaccurate, insufficient, and incomplete data transfer in the integration of BIM into building energy modeling (BEM), thus affecting models' reliability on energy simulations [24]. Murschetz et al. discuss how the inefficient and insufficient collaboration between BIM and BEM systems hampers the quality of data transfer, directly impacting the accuracy of building energy simulation results. They underscore the need for improved interoperability solutions to streamline processes and enhance model quality and compliance in BIM projects [25]. While both publications highlight interoperability issues, Murschetz et al. are centered on sustainability and greening practices, whereas Afzal et al.'s work is focused on improving energy modeling accuracy through better BIM-BEM integration.

By enabling the transposition of IRs into open standard machine-readable language, the IDS provides the foundation for anticipating ACC verification in the design office. This standardization ensures that all stakeholders have a common understanding, thereby facilitating better collaboration and integration across different phases of the project lifecycle [7].

2.4. Automated Compliance Checking

ACC relies on shared rules, valid among stakeholders [13]. For over five decades, attempts have been made to enable the digital verification of standards and regulations [26]. Complex and advanced attempts have been explored to address this issue [27–29], including rule capture [30], data structuring [31,32], and solution development [33]. A preliminary review of a select

number of journals to gather references was conducted. It emerged that 2019 topics related to ACC can be mainly clustered into three groups—safety [34–37], building permit [38–41], and management [42–45]—thus indicating the trends in the scientific community. These studies highlight the need for a more streamlined adoption process, with the IDS offering a potential solution to enhance the efficiency and effectiveness of local compliance checks.

2.5. Information Requirements IDS and Other openBIM Solutions

Clear, well-structured IRs are crucial for successful ACC implementation [46]. IRs must be consistent, reliable, and of uniform quality to ensure effective monitoring throughout all project stages [47–49]. The structuring of IRs according to ACC tools involves the organization of data in a way that aligns with the specific checks and verifications these tools are designed to perform. This alignment is of critical importance for the seamless integration of ACC within the design and approval phases of projects.

The IDS is composed of an XML file that serves a dual purpose: it is designed to be readable by both humans and computers [7]. The specification is divided into two main parts: a description for human comprehension, detailing the title, purpose, and version, and specifications for computer interpretation, including applicability and requirements [50]. These components are further refined into conditions specified by six facets, allowing for the precise definition and filtering of IRs.

The utilization of the IDS in conjunction with other buildingSMART tools, such as the buildingSMART Data Dictionary (bSDD) and the BIM Collaboration Format (BCF), further enriches the modeling process. These tools provide additional semantic context and facilitate communication among stakeholders, ensuring that issues are identified and resolved early in the project lifecycle, as depicted in Figure 1. This automated approach is expected to expedite the design and approval phases while enhancing the project's overall quality and compliance.

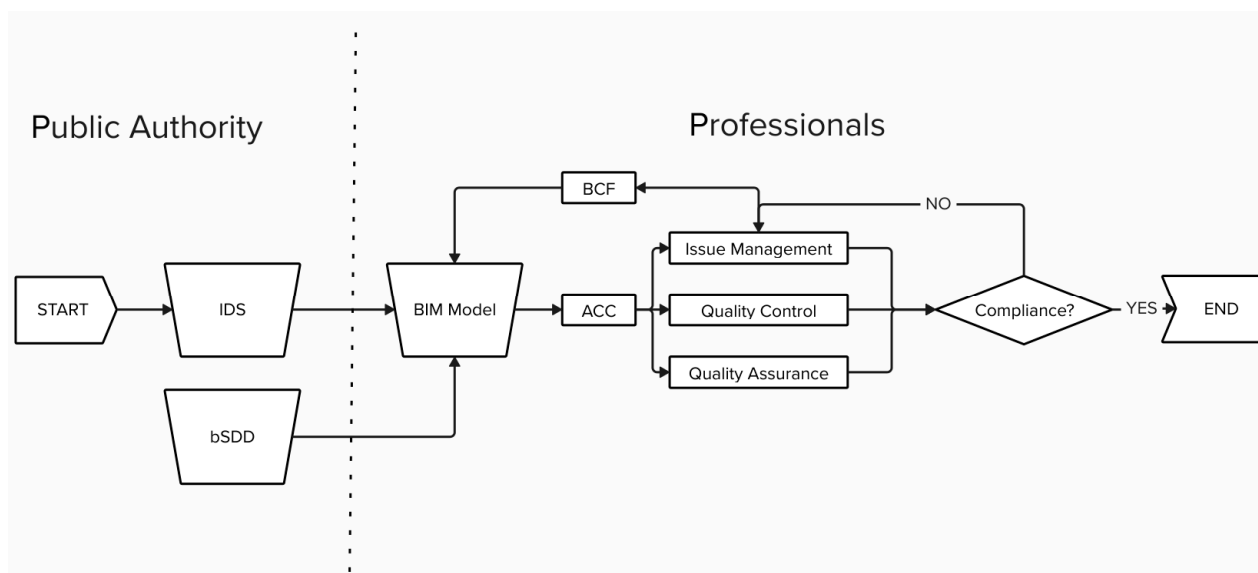


Figure 1. Integration of openBIM solutions within compliance and quality management framework.

2.6. Research Gap and Practical Gap

Currently, the main issue faced by Provincia Autonoma di Bolzano, later referred to as the case study-appointing party, and numerous other public authorities engaged in project tendering is the transfer of IRs to design offices. Professionals involved in public tenders receive administrative and technical documentation, which is mainly PDFs and rarely BIM models in IFC format. This situation implies that information contained in *.pdf format must be manipulated to be transferred to pertinent software (e.g., authoring or checking software) for processing and use. While this process is error-prone and burdensome,

some automation can be implemented through programming. However, this may not be common knowledge among SME design firms. This prompted the question of how to facilitate the automated import of IRs in authoring software while promoting openBIM solutions. In addition, the case study-appointing party might encounter issues related to partially structured BIM models. Despite the existence of established workflows and rules for automated checks, the case study-appointing party was required to adapt and tailor ACC to the specific requirements of each project, based on the IFC file received.

As will be explained later, the IDS offers a solution to some of these challenges by providing a structured framework for specifying and exchanging IRs that, in addition to semantic checks, can address ACC. It is also important to point out that the IDS is a vendor-neutral standard; hence, it is highly suitable for public authorities promoting openBIM approaches.

3. Methodology

3.1. Definition of IRs by Case Study-Appointing Party

As part of the already existing data structure (i.e., a Microsoft® Excel® for Microsoft 365 MSO table and digitalized documents), a novel approach to specifying IRs was developed as an extension of the existing workflow. This novel approach represents a standard way of specifying and exchanging requirements, thus having a modular approach. Further refinement was made to the previously developed data requirements, which were set out in an Excel spreadsheet. This refinement will be explained in the results. The Excel spreadsheet containing the information requirements was structured in 11 groups as property sets and 84 properties and attributes (variously grouped) indicating which properties were relevant for each of the 178 types of objects in all three design stages (the technical and economic feasibility project, final project, and executive project) and facility management. This prevailing information exchange methodology was the only operational framework, and since each design firm developed, to varying degrees, automated property and attribute tagging, the results were still discernible, but at a significant cost in terms of manual labor.

As previously stated, the IDS was proposed as a potential solution to facilitate information exchange. The selection of the IDS standard lies in the opportunity to easily specify and transmit IRs, as well as its structure, which facilitates the alignment of human-readable and machine-readable requirements. To provide an automated import of IRs and compliance check of the requirements, the IRs contained in the Excel spreadsheet were translated into the IDS. This operation had a double scope: first, to update and review the previous data structure, and second, to provide a more IFC-based referencing of objects. To translate IRs into the IDS, the first step was to normalize the information already existing. To do so, each and every case study object, provided with a classification code, was matched with the corresponding *IfcEntity* and further detailed with *IfcTypeObject* (compared in Figure 2). In Figure 2, green cells with the number 1 indicate the required properties while red cells indicate that properties do not apply to an entity. This process facilitated the identification of standard properties and the discernment of any requisite custom properties. This standardized information served as the foundation for the subsequent stage of application development, wherein systematized information was translated into the IDS through the use of *usBIM.IDS* version 1.0 from ACCA Software 1.0, which is one of the many applications available on the market.

Adopted Schema: IFC4X3 (IFC4.3_ADD2)			Classification	7.4.2; 7.5.2; 7.5.3; 7.6.3; 7.9.1; 7.9.2; 9.3.1	1.2.2
			IFC Entity	IfcAudioVisualAppliance	IfcBeam
			IFC Type	AMPLIFIER; CAMERA; COMMUNICATIONTERMINAL; DISPLAY; MICROPHONE; PLAYER; PROJECTOR; RECEIVER; RECORDINGEQUIPMENT; SPEAKER; SWITCHER; TELEPHONE; TUNER	BEAM; JOIST; HOLLOWCORE; LINTEL; SPANDREL; T_BEAM
Property Set	Property Name	Type of parameter			
Identifizierung_Identificazione	Technologische Einheiten_Unità tecnologica	IfcLabel		1	1
Pset_BeamCommon	Name	IfcIdentifier			1
Pset_BeamCommon	IsExternal	IfcBoolean			1
Pset_BeamCommon	LoadBearing	IfcBoolean			1
Pset_BeamCommon	FireRating	IfcLabel			1
Pset_BeamCommon	Status	IfcLabel			1

Figure 2. IfcEntity and standard properties matching according to IRs.

3.2. Information Normalization

Figure 3 illustrates the methodology of the case study objects with the IFC standard, which is a crucial step in integrating building component data into a BIM environment. The process initiates with a case study object, which contains specific attributes and properties related to building components. The initial decision point assesses whether the case study object corresponds to an existing IFC entity. If the object does correspond to an existing IFC entity, it is classified accordingly (e.g., IfcBeam or IfcColumn). This classification enables the object to be associated with standard properties predefined within the IFC schema, thereby enhancing interoperability within various BIM software, ensuring consistency, and addressing data-loss issues for software developers.

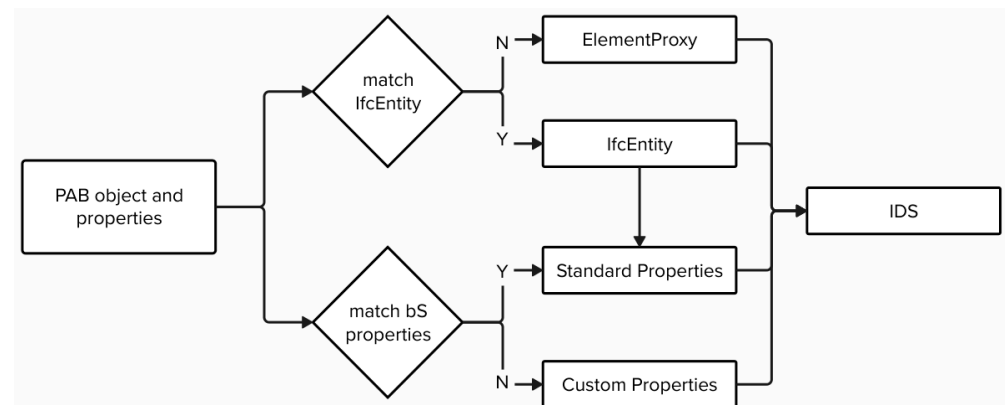


Figure 3. Information normalization workflow conveying to IDS.

If the case study object does not match with an IFC entity, it is designated as an IfcElementProxy. This serves as a placeholder for objects that do not directly correspond to any IFC entity but can still be included in the BIM environment solely with custom properties. For objects classified according to the schema, a subsequent check determines whether they match the buildingSMART properties. If a match is found, the object is then assigned the corresponding standard properties, thereby facilitating further interoperability. If no match is found, the object is assigned custom properties, which are specific to the project or organization and provide additional information not covered by the standard ones.

The final step is to consolidate the objects, regardless of whether they have standard or custom properties, into an IDS.

This bifurcation is crucial as it allows the unique characteristics inherent to specific case study objects while also enabling their integration into the broader BIM ecosystem. It ensures that even the most unique case study objects can be represented within IFC-compliant models, albeit through an adaptable and customized IfcElementProxy. In contrast with the IFC standard properties, custom properties are designed to meet the specific

regional, functional, or design-oriented requirements of the case study. These properties are instrumental in capturing the distinctive characteristics of case study objects, allowing for their detailed representation within the BIM framework, and serving the specific BIM purposes that the appointing party of the case study intends to pursue.

3.3. IDS Collab Tool Design

The development of the mockup tool centered on leveraging the IDS to support both PAs and designers, for modeling information exchange and subsequent verification. The tool was designed to facilitate daily modeling work by ensuring that IRs are consistently applied from the beginning to the end of a project. Based on the activities that professionals must perform, several functions were designed to assist them in their daily activities, see Figure 4. The tool must import the IDS into the authoring software, allowing designers immediate access to all IRs according to the corresponding discipline and design stage. It should enable the reading and interpretation of the IDS for clarity, modeling support, and the mapping of models to align it with the IRs and ensure compliance.

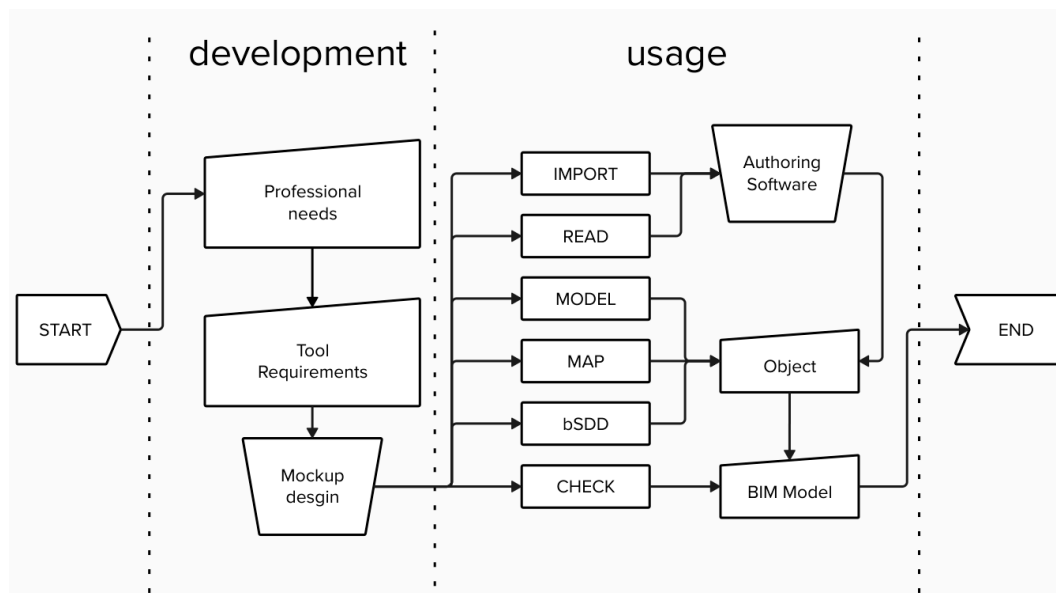


Figure 4. Development and intended uses of IDS Collab Tool.

The mockup of the tool was evaluated through structured questionnaires sent to both the appointing party of the case study and design firms. These questionnaires were designed to gather detailed feedback on the tool's usability, effectiveness, and overall functionality. By collecting insights from both PAs and design professionals, the evaluation assesses the tool's ability to meet diverse user needs and support the entire procurement cycle. This comprehensive feedback informs further refinements, ensuring that the tool is robust, user-friendly, and capable of enhancing efficiency and compliance in BIM modeling and procurement processes. Consequently, the necessity of such a tool is validated.

4. Results

4.1. Case Study Data Structure to IDS

The first significant achievement in this project was the successful translation of the case study IR into the Information Delivery Specification. As mentioned in the Methodology Section, the alignment of existing IRs started with reviewing and updating the previous data structure. The existing data structure was based on UNI 8290:1-1981 (see Table 1). It was organized into 11 disciplines, whose respective categories were not sufficiently able to describe them and, hence, were left blank (i.e., hoisting system and internal equipment).

Table 1. Entities incorporated into case study data structure, according to UNI 8290:1-1981.

Category		Discipline	
Building	38	Structures	7
		Closures	11
		Internal partitions	9
		External partitions	7
		Openings	4
Mechanical	37	Mechanical systems	37
Electrical	48	Electrical systems	48
NA	7	Hoisting system	4
		Internal equipment	3
Firefighting	32	Firefighting system	32
Equipment	16	External equipment	16
178		178	

The 178 entities underwent a matching process according to the architecture of the IFC4X3_ADD2 schema. As shown in Table 2, this resulted in 61 entities belonging to the interoperability layer and its respective elements, 101 entities belonging to the domain layer and its respective domains, and, lastly, 16 entities belonging to the core layer and its respective extension.

Table 2. Number of entities per IFC layer after matching.

Layer		Elements/Domain/Extension	
Interoperability layer	61	Shared building	49
		Shared building service	2
		Shared facilities	6
		Shared infrastructure	2
		Structural element	2
Domain layer	101	Building controls	6
		Electrical	43
		HVAC	43
		Plumbing fire protection	7
		Road	2
Core layer	16	Product extension	16
178		178	

By aligning the existing data with the IDS format, it was ensured that all IRs were accurately represented and seamlessly integrated into the BIM environment. This translation process involved mapping standard and custom IRs to the appropriate IfcEntity attributes, ensuring compliance with both standard industry practices and specific provincial requirements. This step was crucial in establishing a robust foundation for subsequent modeling and verification processes. This result was achieved by leveraging mainly three of the six facets available in the IDS standard. Since all the previously defined objects were matched with IFC classes (e.g., IfcWall, IfcColumn, etc.) and corresponding predefined type (e.g., for IfcWall, elementedwall, movable, parapet, partitioning, plumbingwall, etc., and for IfcColumn, column, pilaster, stand column, etc.), the entity facet was employed to filter elements. Additionally, the property facet was extensively used as the initial spreadsheet

had already revealed the name of the property set, the name of the property, and, to some extent, also the list of expected values. This information made it possible to require a specific class to have a certain property. A similar occurrence was also observed in the case of the attribute facet.

4.2. IDS Collab Tool Mockup

The mockup was designed to demonstrate comprehensive functionality across several key modules, which enhances modeling and review activities for both PAs and design firms. A synthetical explanation of the main functionalities is provided in Table 3 for each of the import, read, model, map, enrich, and check modules.

Table 3. IDS Collab Tool function modules and descriptions.

Function Module	Description
Import	Integrates IDS into the authoring software, ensuring IRs are accessible at the start
Read	Interprets IDS, providing clear, understandable requirements for designers
Model	Supports the creation and modification of BIM models, ensuring they meet IRs
Map	Aligns existing BIM models with IRs, guaranteeing all requirements are addressed
Enrich	Enriches BIM models through bSDD with standardized metadata for utility and compliance
Check	Performs ACC, ensuring the model adheres to all specified requirements

In Figure 5, the software architecture is presented. As previously mentioned, this plugin aims to be integrated into the authoring software and interact with the 3D model, whether it comes directly from the modeler or another external design firm.

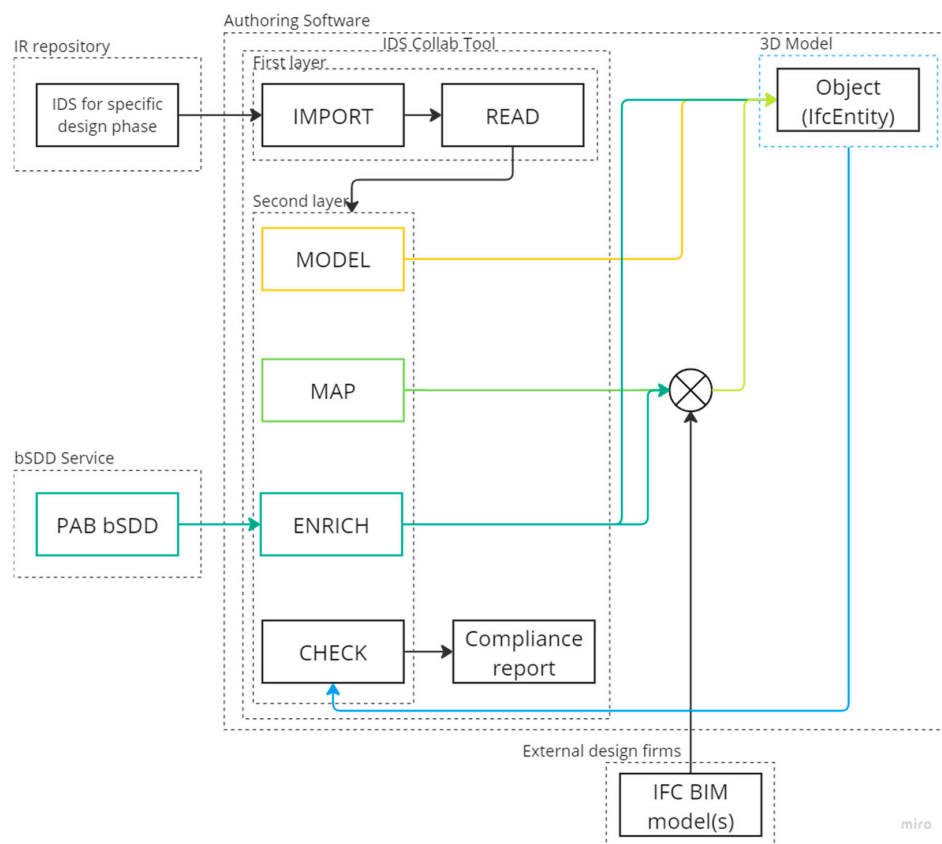


Figure 5. IDS Collab Tool software architecture.

The IDS Collab Tool is organized in two layers, the first one is meant to be preliminary compared with the following one. The first layer allows the designer to retrieve from the IR repository the specific IDS for the design phase. This IR repository can be both provided by the Public Authority and independently stored by each design firm. Once the relevant IDS has been retrieved, the modeler can import the file so that it can be read both by the machine and by the human, based on the dual nature of the IDS. Subsequently, in the second layer, the four main functions are accessible.

The model function is depicted by the yellow box and line. It is intended to facilitate the modeler in the 3D model creation phase. At this stage, the IDS is used by the tool so that all properties and attributes, as required by the public authority, are simultaneously associated with each object created. In this way, each individual object created will already have incorporated all the fields that the designer needs to enhance.

Unlike the previous module, the present one, marked by the light green box, is intended to perform its function with those models coming from outside. In fact, as can be seen in the image below, the map function intersects with the IFC BIM model(s) coming from external design firms to populate each IfcEntity of the BIM model. Similar to the map function, the enrich module, based on a future data dictionary, also intersects with the model coming from external design firms. Unlike its predecessor, however, this module can be integrated with the model created directly in the authoring software from the beginning of design. The association with buildingSMART's service, bSDD, allows the same information model to be available in two different languages, German and Italian, a prerequisite for public procurement in South Tyrol.

The last field is related to the control function. It can use either the IDS previously loaded into the software or any additional IDS file to verify that the information entered corresponds to that required by the customer. This step, which is common to other applications already available on the market, thus makes it possible to verify not only that the requirements have been entered but, at the same time, that they are written in a compliant manner (e.g., to the naming convention agreed upon in the exchange information requirement and the BIM execution plan). The result of this module is the production of a written report and graphical output highlighting the dissimilar objects.

Figure 6 illustrates a screenshot of the mockup tool's functionality after performing a compliance check between the IDS and the BIM model. The tool allows users to efficiently identify and address non-compliant elements. In this example, the tool has highlighted certain elements in yellow, indicating areas that do not meet the specified requirements. The "Result Preview" tab is prominently displayed, listing each highlighted element, such as IfcDuctSegment. For each element, the tool details the specific properties that are non-compliant. For example, the property "WorkingPressure property" is flagged as incorrect. As well as identifying the error, the tool provides a detailed explanation on another tab, specifying the nature of the error and the expected value. This helps users to understand exactly what needs to be corrected.

This feature is crucial for maintaining high standards throughout the modeling process, as it provides real-time feedback to designers. By allowing designers to see which elements are non-compliant and why, the tool facilitates immediate corrections, thereby reducing the likelihood of errors persisting into later stages of the project. This comprehensive checking mechanism ensures that the final BIM model strictly adheres to all specified requirements, enhancing the overall project quality and efficiency.

The results synthesize three main findings: first, the introduction of the IDS plug-in significantly increases workflow efficiency by aligning design processes with the IFC 4x3 schema, thereby promoting greater interoperability and adaptability to designer preferences. Second, the plug-in's real-time compliance checking mechanism ensures that designs comply with predefined standards and regulations, significantly reducing the risk of costly post-design corrections. Third, by incorporating the bSDD, the IDS plug-in standardizes information, improving clarity and accuracy in the exchange of project details across disciplines.

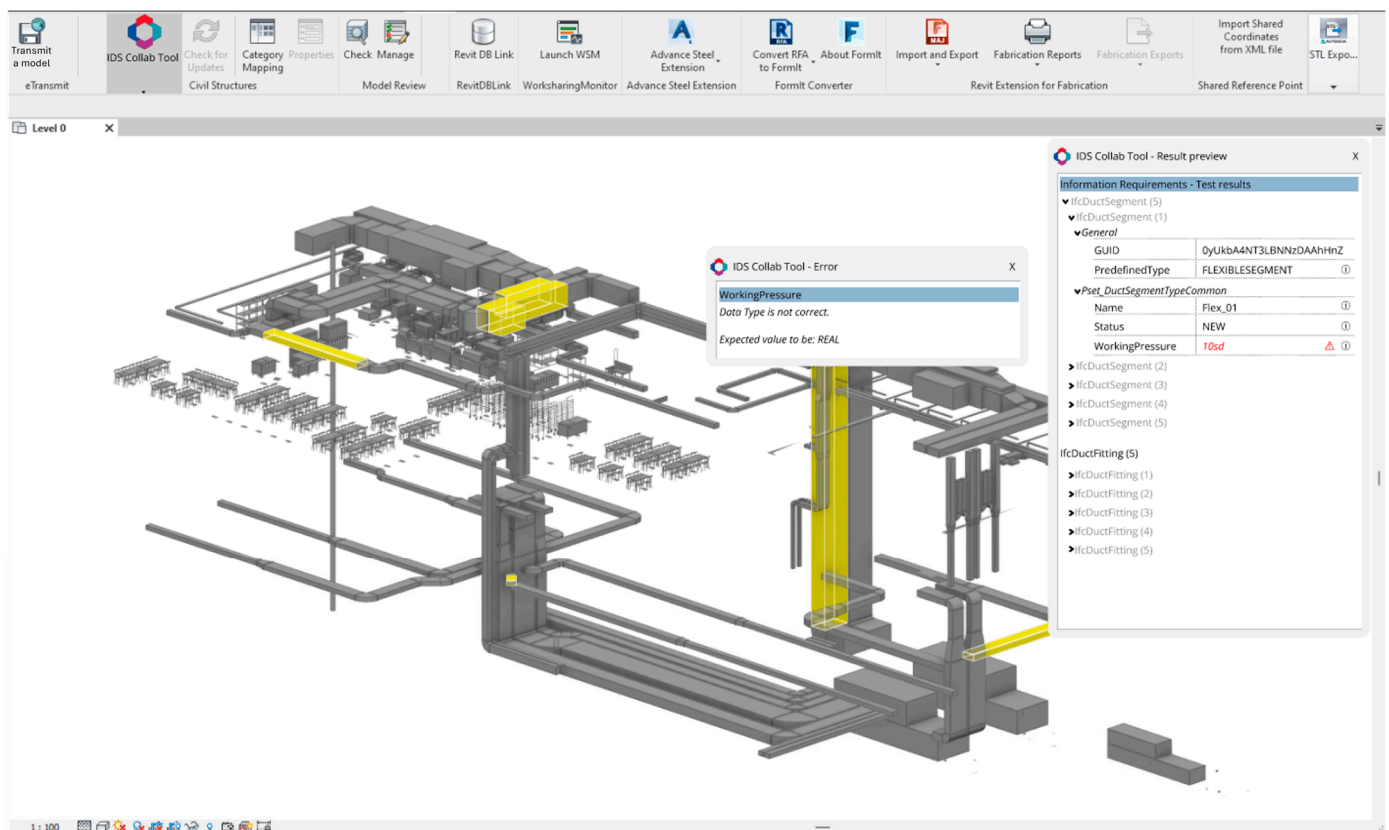


Figure 6. Checked model and corresponding suggestion editing tab.

5. Discussion

This study faced significant challenges in updating properties within the data structure to comply with the latest IFC standard (IFC4X3_ADD2). This involved matching IfcEntity(-ies) with standard and custom properties, ensuring all components and requirements were represented accurately. This effort aligns with findings by Pauwels et al. [14], who demonstrated the potential of semantic rule checking in enhancing consistency and efficiency in validating project requirements. By integrating the IDS, this research aimed to streamline this process, similar to the comprehensive approach to ACC required for universal application across different software platforms highlighted by Eastman et al. [13].

The evaluation through structured questionnaires revealed that 90% of respondents greatly valued the openBIM approach, and 80% of respondents found the tool both highly relevant for the scope of processes in PAs and rated it as useful for workflow efficiency. They rated their IDS knowledge as two on a scale of five. Lastly, 50% of respondents would use it immediately if it was already available. These findings suggest that the IDS, when integrated with a flexible and dynamic tool, holds great promise for optimizing BIM workflows and enhancing project outcomes. This finding parallels the work by Gragnaniello et al. [3], who demonstrated the effectiveness of the IDS in verifying the consistency of ERs but also noted the limited awareness of the standard.

Integrating the novel IDS approach into established industry workflows presents a significant hurdle. Despite the promising capabilities of the IDS plug-in, its full potential remains relatively untapped. The standard is not widely known outside academic circles and in a select group of pioneering companies, indicating a need for broader knowledge and adoption to deep market penetration. This challenge was echoed by Siddiqui et al. [2], who discussed the implementation of new BIM standards to improve interoperability among stakeholders, emphasizing the need for a cultural shift in compliance-checking methodologies. Aligning the capabilities of the IDS plug-in with existing practices will require substantial effort and adaptation.

The introduction of the IDS plug-in significantly increases workflow efficiency by aligning design processes with the IFC 4 × 3 schema, promoting greater interoperability and adaptability to designer preferences. This aligns with the findings by Eastman et al. [13], who outlined the four main steps of the rule-checking process: rule interpretation, building model preparation, rule execution, and rule check reporting. The real-time compliance-checking mechanism of the IDS plug-in ensures that designs comply with predefined standards and regulations, reducing the risk of costly post-design corrections. This is in line with the automated compliance checks discussed by Amor and Dimyadi [26], who highlighted the promise of automated compliance checking in enhancing efficiency and effectiveness in local compliance checks.

The integration of the IDS into building BIM processes represents a significant advancement toward enhancing digital construction methodologies. This study confirms that the IDS can fill a notable gap in current BIM practices by embedding necessary information at an early stage, thereby improving model quality and efficiency. However, the full potential of IDS has yet to be realized due to limited awareness and adoption. Future research should focus on overcoming these barriers and integrating the IDS more seamlessly with existing workflows to fully harness the benefits of digital tools in fostering design and construction practices. This effort requires ongoing collaboration and innovation, as underscored by the broader context of research in this field.

6. Conclusions

The integration of the IDS into building BIM processes, as detailed in this research, highlights a potential advancement toward enhancing digital construction methodologies.

At the core of this study was the assumption that the IDS represents a key innovation for ensuring the provision of accurate, compliant information by designers. This research confirmed that the dual nature of the IDS—as both a standard for specifying IRs and a tool for facilitating its compliance within BIM workflows—addresses a notable gap in current closed BIM practices. In particular, it enhances the design phase by embedding necessary information at an early stage, thereby improving model quality and efficiency regardless of the software vendor.

However, this research also highlighted challenges that need to be addressed. Despite its potential, the IDS standard's full benefits have yet to be realized, mainly due to limited awareness and adoption outside a narrow circle of academia and pioneering firms. Moreover, integrating the novel IDS approach into established industry workflows presents a significant hurdle, requiring a shift in cultural practices within digital construction.

In light of these findings, a call to both academics and industry practitioners to further explore and refine the application of the IDS within BIM processes is made. Future research should focus on overcoming adoption barriers and integrating the IDS more seamlessly with existing workflows. Such efforts are critical for realizing the full benefits of digital tools in revolutionizing design and construction, ensuring that every architectural design achieves precision and compliance.

This study, thus, not only highlighted the transformative potential of the IDS in advancing digital construction practices but also underscored the need for ongoing collaboration and innovation to fully harness this potential.

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